# Lab Four – Programming in R

In this lab, we will build a mealkit recipe generator. I created a small website of about 40 recipes. We will scrape recipes from that website, randomly select three meals, and print a grocery list with recipe cards.

## 1 Preliminaries

I use `R` (Wickham 2025) for all analyses.

### 1.1 Part a

Below, I create a numeric vector filled with random observations. Ask for the 3rd item. **Note**: The `set.seed(7272)` portion ensures we all get the same answer.

```
set.seed(7272) # sets the randomization seed for replication
x <- sample(x=1:10,          # sample from 1, 2, 3, ..., 10
            size=10,         # sample of 10
            replace = TRUE)  # allowed to sample the same item multiple times
```

**Solution**

```
(x[3])
```

```
[1] 5
```

### 1.2 Part b

Below, I create a data frame filled with random observations.

```
set.seed(7272) # sets the randomization seed for replication
df <- data.frame(x1 = sample(x=1:10,          # sample from 1, 2, 3, ..., 10
                             size=10,         # sample of 10
                             replace = TRUE), # allowed to sample the same item multiple times
                 x2 = sample(x=1:10,          # sample from 1, 2, 3, ..., 10
                             size=10,         # sample of 10
                             replace = TRUE), # allowed to sample the same item multiple times
                 x3 = sample(x=1:10,          # sample from 1, 2, 3, ..., 10
                             size=10,         # sample of 10
                             replace = TRUE)  # allowed to sample the same item multiple times,
)
```

### 1.3 Part b

Use the `head(...)` function to peek at the data frame.

**Solution**

```
(head(df))
```

```
  x1 x2 x3
1  8  1  8
2  3  7  2
3  5  3  3
4  3  6  2
```

```
5  2  2  3
6  3  8 10
```

## 1.4 Part c

Ask for the column `x1`.

**Solution**

```r
(df$x1)
```

```
 [1] 8 3 5 3 2 3 1 2 6 2
```

## 1.5 Part d

Ask for the fifth row of the data frame.

**Solution**

```r
(fifth.row <- (df[5,]))
```

```
  x1 x2 x3
5  2  2  3
```

## 1.6 Part e

Ask for the the value of `x1` in the fifth row of the data frame.

**Solution**

```r
(fifth.row$x1)
```

```
[1] 2
```

## 1.7 Part f

Ask for the the value of in the third column and the fifth row of the data frame.

**Solution**

```r
((df[5,])$x3)
```

```
[1] 3
```

## 1.8 Part g

Create a sequence from 1 to 10 by 2 and use it to print the odd rows of the data frame.

```r
(sequence <- seq(from=1, to=10, by=2))
```

```
[1] 1 3 5 7 9
```

```r
(df[sequence,])
```

```
  x1 x2 x3
1  8  1  8
3  5  3  3
5  2  2  3
7  1 10  1
9  6  2 10
```

## 1.9 Part h

Below, I create an empty column called `x12`. Fill in the details of the `for(...)` loop to ensure `x12` is the product of `x1` and `x2`.

**Solution**

```
n <- nrow(df)                # How many rows to we have to fill?
df$x12 <- rep(NA, nrow(df))   # Create an empty column for x12

for(i in 1:nrow(df)){
  x1 = df[i,]$x1
  x2 = df[i,]$x2
  df$x12[i] = (x1*x2)
}
(df$x12)
```

```
 [1]  8 21 15 18  4 24 10 12 12  8
```

## 1.10 Part i

Write a function called `calculate.score(...)` that takes three arguments representing `x1`, `x2`, and `x3`) and returns a single value based on the formula:

$$Score = (x_1 \times 2) + x_2 - x_3$$

Use your function to create a new column `total.score` in our data frame `df`.

**Solution**

```
calculate.score = function(x1,x2,x3) {
  (x1*2)+x2-x3
}
df$total.score <- rep(NA, nrow(df))
  for(i in 1:nrow(df)){
  x1 = df[i,]$x1
  x2 = df[i,]$x2
  x3 = df[i,]$x3
  df$total.score[i] = calculate.score(x1,x2,x3)
}
(df$total.score)
```

```
 [1]  9 11 10 10  3  4 11  9  4  7
```

## 1.11 Part j

Create a function called `evaluate.row(...)` that takes one argument and returns "low" when the argument is less than 4, "mid" when the argument is between 4 and 7 (inclusive), and "high" when the arguement is 8 or larger. Then, use `sapply(...)` to apply it to column `x1`.

**Solution**

```
evaluate.row = function(x){
  if(x < 4) {
    return("low")
  } else if (x >= 4 & x <= 7) {
    return("mid")
  } else {
    return("high")
  }
}
sapply(df$x1, evaluate.row)
```

```
 [1] "high" "low"  "mid"  "low"  "low"  "low"  "low"  "low"  "mid"  "low"
```

### 1.12 Part k

Did we need to use loops or functions in (h.)-(j.)? That is, can we use vectorization to attain the same results in 1 line each? Where it is possible, write the line of code. Where it is not, explain why.

**Solution**

We can use vectorization for h and i.

```
df$x12 = (df$x1)*(df$x2)
df$total.score = ((df$x1)*2+df$x2-df$x3)
```

We can't use vectorization in part j because the block notation of `if()` function are not vectorized (it doesn't work simultaneously on all elements of a vector, but process each element at once to return TRUE/FALSE). However, we can use the vectorized `ifelse` function:

```
df$category <- ifelse(df$x1 < 4, "low", ifelse(df$x1 <= 7, "mid", "high"))
```

## 2 Complete Tasks for One Recipe

### 2.1 Part a

Install and load the `rvest` package (Wickham 2025).

**Solution**

```
# install.packages("rvest")
library("rvest")
```

### 2.2 Part b

Load the html of the `website/KimchiGrilledCheese.html` using the `read_html()` function. We will use the Kimchi Grilled Cheese recipe as our prototype and extend this workflow to all recipes, so try to be as general as possible. If you do look at it, you'll notice it contains the html we saw in the developer tools in class. **Hint**: You can use `read_html(...)` like `read.csv(...)`. Don't forget you can use the documentation to help use it.

**Solution**

```
recipe.item <- read_html("website/KimchiGrilledCheese.html")
```

### 2.3 Part c

Open the html file in a web browser and open developer tools. In chrome-based browsers, you can do this by pressing the three verticle dots in the upper-right corner, clicking "more tools", then "developer tools".

Find the name of the Ingredients section of the website and pull the HTML of the function using `html_element(...)` and save the results to an object called `ingredients.section`. **Hint**: You can ask for elements by id using a preceeding "#". See `?html_element(...)` for a helpful example.

**Solution**

```
(ingredients.section = html_element(recipe.item, "#ingredients"))
```

```
{html_node}
<section id="ingredients" class="level2">
[1] <h2 class="anchored" data-anchor-id="ingredients">Ingredients</h2>
[2] <ul>\n<li>1/8 tsp chili flakes</li>\n<li>4 garlic cloves</li>\n<li>5 oz s ...
```

### 2.4 Part d

Now, we want to obtain all of the itemized items. Find the element type the individual ingredients and pull all of them from `ingredients.section` using `html_elements(...)` (note the added s) and save the results to an object

called `ingredients`. **Hint**: You can ask for elements by type by simply specifying the tag (e.g., "p" for paragraph). See '?html_element for a helpful example.

**Solution**

```
(ingredients = html_elements(ingredients.section, "li"))
```

```
{xml_nodeset (9)}
[1] <li>1/8 tsp chili flakes</li>
[2] <li>4 garlic cloves</li>
[3] <li>5 oz spinach</li>
[4] <li>1 tsp gochujang</li>
[5] <li>3 tsp mayonnaise</li>
[6] <li>2/3 cup kimchi</li>
[7] <li>4 slices of bread</li>
[8] <li>2/3 cup cheddar cheese</li>
[9] <li>4 tablespoons everything seasoning.</li>
```

## 2.5 Part e

Similar to Part c. Find the name of the Instructions section of the website and pull the HTML of the function using `html_element(...)` and save the results to an object called `instructions.section`.

**Solution**

```
(instruction.section = html_elements(recipe.item, "#instructions"))
```

```
{xml_nodeset (1)}
[1] <section id="instructions" class="level2"><h2 class="anchored" data-ancho ...
```

## 2.6 Part f

Now, we want to obtain all of the enumerated items. Find the element type the individual instructions and pull all of them from `instructions.section` using `html_elements(...)` (note the added s) and save the results to an object called `instructions`.

**Solution**

```
(instructions = html_elements(instruction.section, "li"))
```

```
{xml_nodeset (5)}
[1] <li>In a frying pan, heat 2 tablespoons olive oil with chili flakes.</li>
[2] <li>Add spinach and cook until wilted.</li>
[3] <li>Mix gochujang and mayonnaise in a small bowl and spread on one side o ...
[4] <li>Build sandwiches by layering spinach, kimchi, and cheese between two  ...
[5] <li>Fry in butter until golden brown on both sides.</li>
```

## 2.7 Part g

Find the class of the recipe image and pull the HTML using `html_element(...)` and save the results to an object called `image.element`. **Hint**: You can ask for elements by class using a preceeding ".". See `?html_element(...)` for a helpful example. Further, note that HTML elements may have more than one class (separated by a space). When that is the case, you need to choose one.

**Solution**

```
image.element = html_element(recipe.item, ".figure-img")
```

## 2.8 Part h

Use the `html_attr(...)` function to pull the source link ("src"). Then, use `paste(...)` to prepend the source link with "website/" so we have the full link. **Note**: This would be like adding "https://www.website.com" to get the

absolute link.

**Solution**

```
image.url = paste("website/",html_attr(image.element, "src"), sep="")
```

## 2.9 Part i

Now that we have all the things we need, let's try to print the recipe. Below, I have written code to print from the objects. One by one, remove `#| eval: false` from the YAML header and add `#| echo: false` and `#| results: 'asis'`, and test. Let me know if you're stuck!

**Image**

Figure 1: Kimchi Grilled Cheese

**Ingredients**

- 1/8 tsp chili flakes
- 4 garlic cloves
- 5 oz spinach
- 1 tsp gochujang
- 3 tsp mayonnaise
- 2/3 cup kimchi
- 4 slices of bread
- 2/3 cup cheddar cheese
- 4 tablespoons everything seasoning.

**Instructions**

1. In a frying pan, heat 2 tablespoons olive oil with chili flakes.
2. Add spinach and cook until wilted.
3. Mix gochujang and mayonnaise in a small bowl and spread on one side of each slice of bread. Sprinkle with everything seasoning and press it into the bread.
4. Build sandwiches by layering spinach, kimchi, and cheese between two slices of bread.
5. Fry in butter until golden brown on both sides.

# 3 Complete a Full Menu!

Open `Menu.qmd` and add code to complete the following.

1. Randomly select three dinner recipes and one breakfast recipe at random. **Hint:** Use the `sample(...)` function.
2. Pull the image, ingredients, and instructions for each selected recipe.
3. Combine all of the ingredients into a grocery list on the first page.
4. Print the image, ingredients, and instructions for each receipe on the subsequent pages.

When you render this document, no code should be visible. Instead, you should see a five-page document as described above.

# 4 Describe your work!

## 4.1 Why a fake website?

Read https://www.scrapingbee.com/blog/is-web-scraping-legal/. Originally, I conceived doing this with recipes from Purple Carrot (my favorite subscription service). Being a large company, their terms of service are *very* long and precluded us from copying their recipes. We also checked a few smaller recipe websites we like and even they had terms of service that restricted automated collection of data.

## 4.2 Conditionals

Did you use conditional statements in your code? If yes, how. If not, are there places you could have used them but did something else?

**Solution**

I didn't use conditional statements in my code. I don't think there are places where I could have used them, unless some websites were missing some elements (e.g., missing title or the image) and in that case maybe it's helpful to give a warning.

## 4.3 Loops

Did you use loops in your code? If yes, how. If not, are there places you could have used them but did something else?

**Solution**

I used loops in my code to pull the ingredients, instructions and image URLs for all 4 recipes then store them in 3 lists, instead of doing one recipe at a time. This was really helpful for the task in which I have to print all of the ingredient items. I could also make use of the lists storing the data from the loop to simplify my codes to pull image, ingredients, and instructions for each selected recipe.

## 4.4 Functions

Did you use functions in your code? If yes, how. If not, are there places you could have used them but did something else?

**Solution**

I could have used a function to produce a list of all elements required for each recipe. It may look something like this:

```
link = as.character(df$recipe.link)
recipe.info <- function(link){
  recipe.item <- read_html(link)
  ingredient.items = html_elements(html_element(recipe.item, "#ingredients"), "li")
  instructions = html_elements(html_elements(recipe.item, "#instructions"), "li")
  image.url = paste("website/",html_attr(html_element(recipe.item, ".figure-img"), "src"), sep="")
```

```
return(list(ingredient.items,instructions,image.url))
}
```

Then for the list of all ingredients, I can combine the `ingredient.items` element of each list generated for each recipe.

## References

Wickham, Hadley. 2025. *Rvest: Easily Harvest (Scrape) Web Pages*. https://doi.org/10.32614/CRAN.package.rvest.